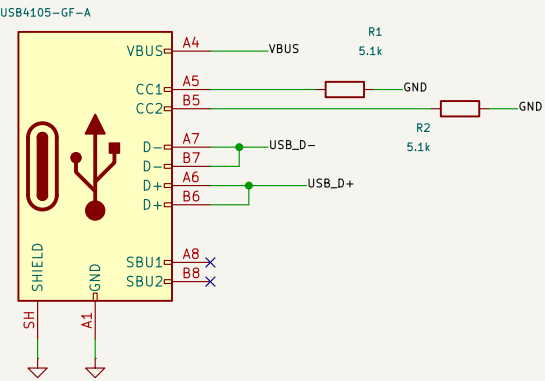


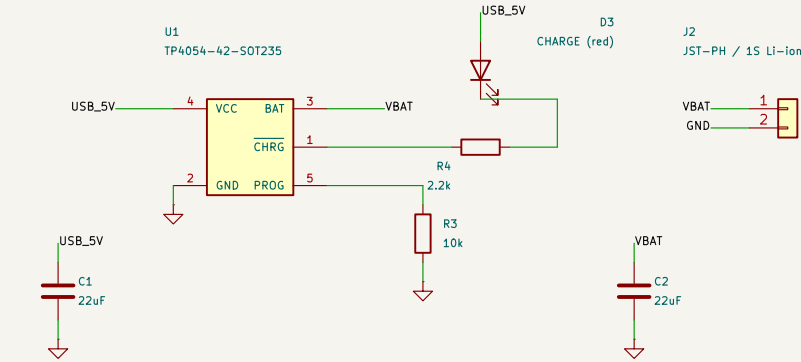
NUCULA v2 / USB-C, Li-ion power and ESP32-C3

01 USB 2.0 device / 5 V sink



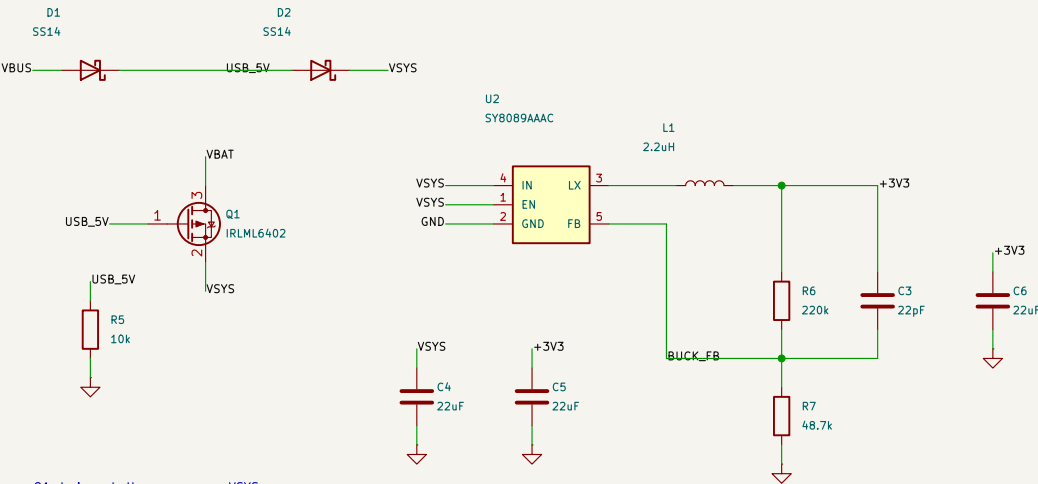
CC1 / CC2: independent 5.1k Rd. No USB-PD.
Route D+/D- as a 90 ohm differential pair.
Place U4 at connector; R12/R13 at module.
See design notes for USB current / inrush limits.

02 4.2 V / 100 mA charger



Protected 1S LiPo, ~400mAh (3.7V nom / 4.2V full).
J2: JST-PH 2.0mm; pin 1 = BAT+; pin 2 = GND.
100mA charge = 0.25C at 400mAh; ~10mA termination.
Confirm pack polarity and ratings; no cell-temp input.

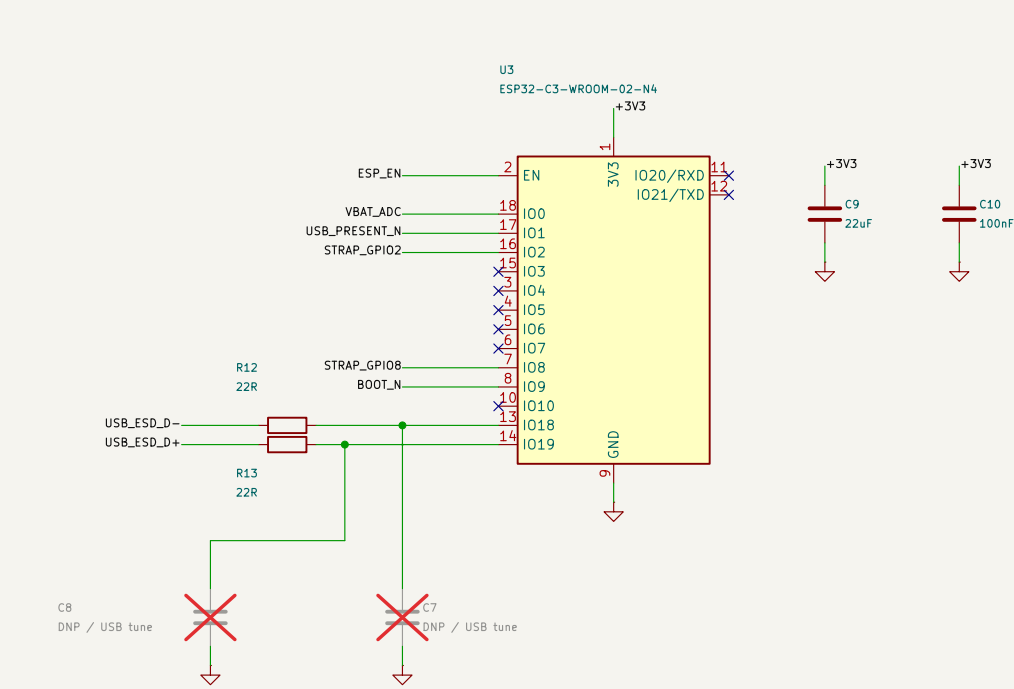
03 USB-priority supply / 3.31 V buck



Q1 drain = battery, source = VSYS.
Body diode points VBAT -> VSYS.
USB load bypasses charge-current sensing.

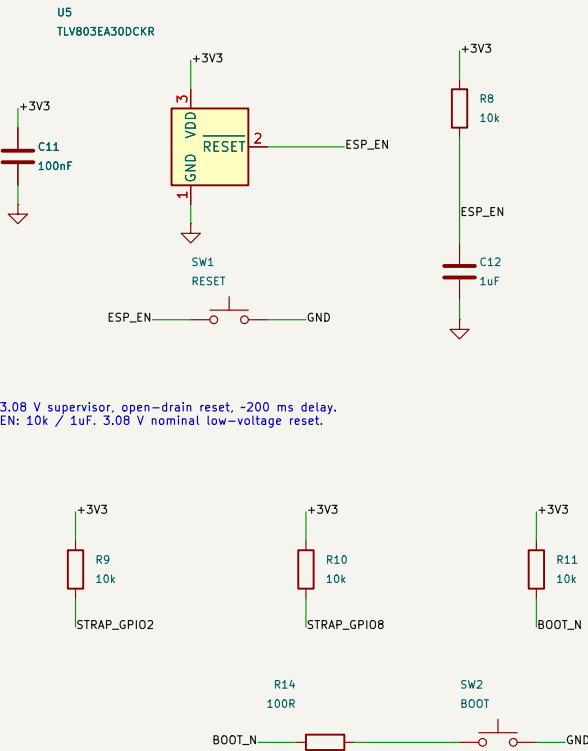
VOUT = 3.31 V; R6 / R7: 0.1% precision.
500 mA rail design targets validate transients.
Buck only: low battery causes dropout / reset.
C5 + C6: 44 uF nominal; verify DC-bias derating.

04 ESP32-C3-WROOM-02-N4 / native USB



USB D- = GPIO18 (pin 13); D+ = GPIO19 (pin 14).
No UART bridge. Unused GPIOs deliberately marked NC.
Keep antenna clear; bond EP pin 19 and pin 9 to ground.
Use ADC1 GPIO0 for battery sensing (not ADC2).

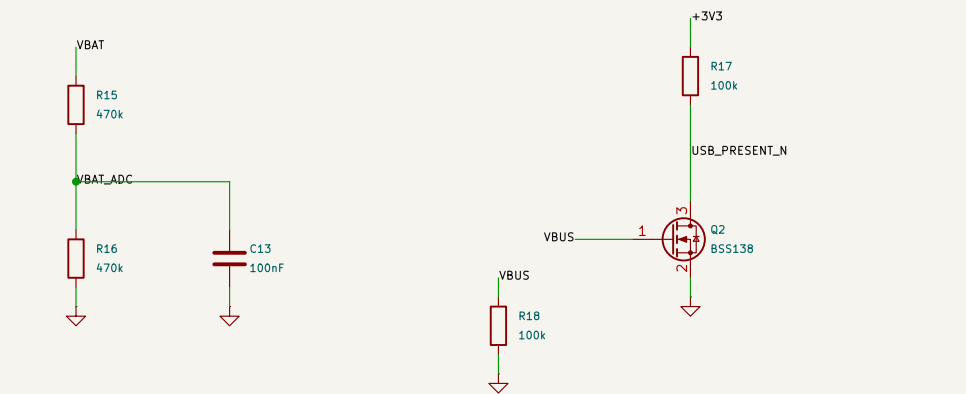
05 Reset, undervoltage and boot straps



3.08 V supervisor, open-drain reset, ~200 ms delay.
EN: 10k / 1uF, 3.08 V nominal low-voltage reset.

Hold BOOT; tap RESET to enter ROM download mode.
GPIO8 must stay high while GPIO9 is low.
No large capacitor on GPIO9.

06 Battery and USB presence sensing



VBAT_ADC = VBAT / 2, max ~2.15 V incl. tolerances.
ADC1 GPIO0: use calibrated 11/12 dB attenuation.
Divider load ~4.5 uA; allow >=120 ms to settle.

USB_PRESENT_N: GPIO1 LOW when VBUS is present.
Firmware must disable USB attach when VBUS is absent.
Q2 isolates 5 V VBUS from the 3.3 V GPIO supply.

DRAFT / before PCB:
Confirm protected battery and JST polarity.
Validate USB current, inrush, suspend and ESD.
Verify regulator layout, capacitor bias and load steps.
Reference: Olimex ESP32-C3-DevKit-Lipo Rev C.
See docs/power-design.md for sources and limits.

Schematic only / protected 4.2 V single-cell Li-ion / 100 mA charge
Adapted from Olimex ESP32-C3-DevKit-Lipo Rev C; CERN-OHL-S-2.0

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File: nucula-v2.kicad_sch
Title: Nucula v2 - USB-C, battery power and ESP32-C3

Size: A3 Date: 2026-09-20 Rev: A-draft
KiCad E.D.A. 10.0.6 Id: 1/1

ERC source declarations